Temporal update and spatial reference studies

2026-10-14 | TA-STVG + frozen STVG-R1-7B | fixed historical development cohort: HC2 4, VidSTG 4. All results are finite pilot measurements, not confirmation or a new deployed method.

P1/P2: stable optimization does not imply correct localization

Temporal readout / frozen boxes	HC vIoU	HC tIoU	Vid vIoU	Vid tIoU
Native / frozen	22.210	46.415	14.490	27.778
Projection / zero residual	27.851	58.434	22.016	40.807
Original Adam	31.286	60.486	22.442	40.905
One-shot ridge	23.079	53.040	23.187	42.825
Five-round residual ridge	22.422	52.074	23.187	42.825
Ridge + full-span trust	22.214	49.627	23.187	42.825
Per-offset target + geometry	22.554	49.316	23.187	42.825

40 cached temporal fits; no additional teacher calls or model forwards. Same 512-parameter interface, tau 1 and L2 .001. Ridge targets centered rewards; trust accepts actual steps only when each full legal-span reverse KL <= .01 and the fixed objective does not increase. Original Adam exactly reproduces the prior six-snapshot trace.

Original Adam fixed objective increased in 7/8 cases and final parameter component outside evidence subspace reached .542718. Low-rank modes stayed within 1.71e-14; trust maximum full-KL .00999637, per-offset .00999684. Mean feature rank 2-3; stacked offset rank 3-6.

P2 attribution boundary: per-offset targets/KL and separately centered stacked feature geometry changed jointly. This test cannot isolate their individual contributions. The pre-execution protocol and predictions were retained; correction is recorded separately.

P3: same reference, actual student re-decode

8 new Direct fits and 8 new source-capped OPD fits, five steps each; 8 actual frozen source captures and 16 actual final full-model readouts. Source pixels/checkpoint/frame grid preserved; final full-model boxes exactly equal cached decoder outputs. Gaussian formulas and old domain configurations are retained.

Spatial arm	HC Native vIoU	Vid Native vIoU	HC + trust T	Vid + trust T
Frozen source	22.210	14.490	22.214	23.187
Direct control	21.556	14.490	21.698	23.187
Old Gaussian OPD (saved)	21.213	13.513	21.613	21.881
Source-capped Gaussian OPD	21.983	14.702	22.053	23.490

40/40 spatial proposals accepted after 184 backtracking attempts. All 14,160 source-frame box coordinates satisfy per-frame normalized cxcywh RMS<=.01. No actual all-rejected case occurred; exact restore branch has source/CPU-mock evidence, not an exercised GPU failure path.

Unobserved dense GT frames: sIoU delta (pp)	HC2	VidSTG
Direct	-1.122	0.000
Old OPD	-0.208	-2.801
Source-capped OPD	-0.364	0.499

The sparse reference is a teacher-selected student pseudo-reference, not independent teacher geometry. Protecting a wrong source box can preserve its error; box displacement protection does not guarantee unobserved-frame benefit. Direct is an authorized diagnostic control, not a mandatory paper benchmark.

Supervision direction and all-case accounting

Offline diagnostic only: no GT gradient, score, best step or oracle was used for any update, candidate, backtracking acceptance or hyperparameter. All new predictions were sealed before postseal metric/gradient readback.

Case	QA gradient vs GT gradient	Ridge delta vs GT descent	Per-offset delta vs GT descent	Conservative S delta vIoU (pp)
HC01	0.047	0.634	0.414	-0.012
HC02	-0.764	-0.690	-0.346	-1.076
HC03	-0.329	-0.349	-0.050	0.090
HC04	-0.059	-0.595	-0.360	0.088
V01	0.872	0.513	0.514	0.611
V02	-0.536	-0.364	-0.383	0.043
V03	-0.005	0.200	0.087	-0.027
V04	0.092	0.147	0.148	0.220

Low-rank first-step cosine to offline GT descent: HC2 -.249834 / VidSTG +.124126; per-offset -.085785 / +.091329. Improved optimization stability has not fixed the supervision direction in both domains. 304 official/independent metric readbacks agree to 1.23e-15. All positive, negative, neutral and unknown records are published in the anonymous scalar report.

Decision: no fixed two-domain replacement is selected from this pilot; no 32-parent-per-direction confirmation, carry experiment, old queue restoration or CURRENT promotion was triggered. This closes the optimizer/reference comparison, not the overall MLLM research route.

Box-scale global evidence: new P0 qualification

This separately executed spatial proposal searches beyond the student local candidate pool. One fixed anchor per historical source, full coarse same-scale grid; two frozen R1 scoring protocols: referent-specific blur response and marked-window containment. One frame is a new input-protocol boundary; no full Tube or re-identification claim.

Five of eight prelocked anchors have saved GT boxes (HC2 3, VidSTG 2); three remain unknown without GT-informed anchor replacement. All 184 categorical calls are sealed, including 24 separately declared source-position controls.

Case	Grid windows	Source IoU	Oracle IoU	Mask-selected IoU	Mark-selected IoU
HC01	9	0.589	0.577	0.000	0.493
HC02	12	0.579	0.452	0.000	0.000
HC03	4	unknown	unknown	unknown	unknown
HC04	21	0.054	0.350	0.150	0.000
V01	12	0.672	0.492	0.000	0.492
V02	4	0.653	0.731	0.731	0.416
V03	6	unknown	unknown	unknown	unknown
V04	8	unknown	unknown	unknown	unknown

IoU values are fractions. Grid coordinates come only from source width/height; oracle is diagnostic after sealing and never enters teacher prompts or selection. Mask selection maximizes original support log-odds minus blurred-window support log-odds; marked selection maximizes referent containment support. Attribution response is not a calibrated localization probability.

76 complete coarse windows and 160 fixed single-image categorical requests; no half-step refinement, multiscale extension, DINO/YOLO/SAM2, student fit or Gaussian policy update in this P0. All grid candidates and feedback, including negative/flat/unknown results, are retained.

Global grid + unchanged source-position control

The complete coarse grid can miss the original student center. A separately declared control after first-P0 readback added 8 source-mask, 8 source-mark and 8 unchanged-image repeats, with identical prompts and no GT-derived coordinates. Original 160 records are immutable. All 8 repeated-image logits/probabilities/log-odds match exactly across teacher loads; no prompt search or new anchor selection.

Case	GT anchor	Union oracle IoU	Mask + source IoU	Mark + source IoU
HC01	available	0.589	0.000	0.493
HC02	available	0.579	0.579	0.579
HC03	unknown	unknown	unknown	unknown
HC04	available	0.350	0.150	0.000
V01	available	0.672	0.000	0.492
V02	available	0.731	0.731	0.416
V03	unknown	unknown	unknown	unknown
V04	unknown	unknown	unknown	unknown

Valid-anchor mean Source / union oracle / Mask / Mark: HC2 .407344 / .506033 / .242903 / .357375; VidSTG .662731 / .701445 / .365303 / .453882. Allowing the source candidate improves the comparison, but teacher selection still loses mean IoU in both directions. Mask selection retains 2 positive, 2 negative and1 neutral cases; Mark has 0 positive, 4 negative and1 neutral. Three anchors stay unknown.

No spatial parameter adaptation or complete-method expansion was triggered for this fixed single-anchor recipe. This negative qualification is not a universal rejection of global evidence or frozen R1. Coarse same-scale limits and one-frame referent/relationship ambiguity remain separate hypotheses.

CPU-only 32-sample matched local-Gaussian reward probes: 2/16 reference/protocol pairs have exactly zero IoU rewards; all DIoU rewards vary. This validates reward observability only, not the correctness of the selected target or learned localization.

Measured execution: 160 original model-forward+sync calls 17.346178s, teacher worker wall 56.103196s; 24 control calls 2.977674s, worker wall 28.779893s. Image preparation and full per-request latency were not timed and remain unknown. Peak allocated GPU 16,735,618,560bytes; effective owned-unit RAM high/max/swap 16/20/1GiB checked. No observed OOM, new downloads, DINO, YOLO or SAM2.

References: RISE, arXiv 1806.07421; D-RISE, CVPR 2021 pp 11443-11452; DIoU, arXiv 1911.08287. These motivate attribution/geometry diagnostics; they do not establish MLLM localization accuracy.